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C499 Computer Science Capstone
5-2 Milestone Four: Enhancement Three, Databases
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Enhancement Three: Databases

For this milestone, I enhanced the Fire Department Event Tracker application by improving how event information is stored, managed, and maintained within the SQLite database. The purpose of this enhancement was to demonstrate knowledge and practical application of database design, schema modification, data persistence, and database integration within a mobile application environment. These improvements strengthened the application's ability to manage event records while providing a more realistic and maintainable solution for long-term event tracking.

The original artifact allowed users to create, edit, and delete events using SQLite database storage. Event records contained information such as the event title, date, time, and location. While the original implementation successfully stored and displayed event information, it lacked a mechanism for managing the lifecycle of events after they had been completed. Once an event was no longer needed, the only available option was to delete the record from the database entirely.

To improve the functionality of the artifact, I implemented event lifecycle tracking through a database schema enhancement. A new status field was added to the SQLite events table, allowing event records to be categorized as Upcoming, Completed, or Archived. This enhancement allows records to remain in the database while reflecting their current status rather than requiring permanent deletion. By preserving event records, the application provides improved data management and creates opportunities for future reporting and historical record keeping.

Implementing this enhancement required several modifications throughout the application. The SQLite database schema was updated to include the new status column, and the database version number was increased to support the schema change. Database helper methods were updated to read and write status values, while the Event model was expanded to store status information as part of each event object. These changes ensured that status values could be persisted within the database and accessed throughout the application.

In addition to modifying the database structure, I integrated status management directly into the user interface. The RecyclerView event list was updated to display the status associated with each event record. A long-press action was also implemented, allowing users to update event status values through a dialog interface. This enhancement improved usability by providing a simple and efficient method for managing event records while maintaining synchronization between the user interface and the underlying SQLite database.

Additional improvements included updating data retrieval operations, modifying event object construction, and improving overall maintainability through reusable database methods. Rather than creating separate methods for each status type, a single reusable update method was implemented to manage all event status changes. This approach reduced duplicate code and improved long-term maintainability of the application.

One challenge encountered during development involved modifying the existing database schema while maintaining compatibility with previously stored event records. Updating the database version and ensuring that all components of the application correctly recognized the new status field required careful testing and validation. Another challenge involved ensuring that status changes were properly reflected throughout the application, including the SQLite database,

Event model, RecyclerView adapter, and user interface components. These issues were resolved through incremental testing and consistent updates across all affected classes.

Overall, this enhancement strengthened the application by improving database management, data persistence, and event lifecycle tracking while demonstrating practical implementation of database design principles, schema evolution, CRUD operations, and mobile database integration. While completing this enhancement, I realized that a future improvement for the application would be implementing automatic archival of expired events once their scheduled dates have passed. As event records continue to accumulate over time, automated lifecycle management could further improve long-term organization and efficiency. This is a feature I would like to explore after graduation as I continue expanding my software development skills and enhancing the application.